Technical note: ASTRA weighted correlation functions — replacing environment *splits* with a continuous *weight*

ASTRA clustering project

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Abstract

The ASTRA analysis classifies each object by a local cosmic-web parameter $r = (n_{\mathrm{data}} - n_{\mathrm{rand}})/(n_{\mathrm{data}}+n_{\mathrm{rand}}) \in [-1, 1]$ obtained from a Delaunay tessellation over the joint data+random catalog, then splits the sample into quantiles of r and measures a two-point correlation function (2PCF) per quantile. Here we develop an alternative that uses r directly as a per-object *weight* and measures *weighted* (marked) correlation functions, avoiding the information loss and bin-edge discontinuities of the split. We are careful to keep three populations distinct: the *data* galaxies (which carry redshift-space distortions), the *ASTRA randoms* (the 1× uniform points that enter the Delaunay and therefore carry an r), and the *geometry randoms* (the 5× uniform catalog that is only the Landy–Szalay reference). Crucially, only the two r-carrying populations are weighted; the geometry randoms define the window and are never weighted. We present the estimator (a unified weighted Landy–Szalay with mean-normalised weights, so the uniform weight reproduces the standard 2PCF exactly), seven weight schemes, three statistics (data auto, ASTRA-random auto, and their cross), and the first full-box measurement on the fiducial `c000/hod484` box. This note documents the method in full; the results section is populated as the first run completes.

1 Motivation

The quantile split has two practical drawbacks. First, it discards information: the continuous r is reduced to a four-level label, and the bin edges are arbitrary. Second, it is not smooth: the bin boundaries introduce discontinuities that are awkward for the emulator/Fisher derivatives that the rest of the project relies on, and they enlarge the data vector (four quantiles × several leg families), inflating the Hartlap penalty. A continuous weight $w = f(r)$ keeps the full r information, is differentiable in the cosmological and HOD parameters, and produces a compact data vector. Weighted ("marked") correlation functions are an established tool in large-scale structure with known sensitivity to modified gravity, massive neutrinos, and assembly bias [1, 2, 3], which strengthens the science case.

2 Three populations, and which ones are weighted

The pipeline keeps three populations strictly separate; conflating them is the main pitfall.

- **Data** (N_{d} galaxies): carry redshift-space distortions (the line-of-sight coordinate is `Z_RSD`); each has an r.
- **ASTRA randoms** (1× data; regenerated each iteration with seed `SEED+iter`): the uniform points that enter the Delaunay tessellation, so each carries an r. These are the "random quantile" tracers of the split analysis.

- **Geometry randoms** ($5\times$ data; fixed seed): a uniform catalog used *only* as the Landy–Szalay reference (the survey window). They do *not* enter the Delaunay and therefore have no r.

Because the geometry randoms have no r, they cannot and must not be weighted by $f(r)$: weighting the LS reference by a data-derived quantity would distort the window rather than the signal. The weights attach only to the data and the ASTRA randoms; the geometry randoms remain uniform and unweighted. This is the continuous analog of the split analysis, in which the data and the ASTRA randoms are each binned, while the geometry randoms provide the common reference for every 2PCF.

3 Estimator

3.1 The environment parameter

For each object in the joint catalog, the Delaunay neighbour counts give

$$r = \frac{n_{\text{data}} - n_{\text{rand}}}{n_{\text{data}} + n_{\text{rand}}} \in [-1, 1], \tag{1}$$

with $r \to -1$ in voids and $r \to +1$ in knots. Objects with no neighbours ($n_{\text{data}} + n_{\text{rand}} = 0$, exceedingly rare) have r undefined and are dropped uniformly across all schemes, so every scheme uses the same sample.

3.2 Weighted Landy–Szalay with mean-normalised weights

Given a per-object mark $m_i = f(r_i)$ for a tracer population, we form mean-normalised weights

$$w_i = \frac{f(r_i)}{\langle f(r) \rangle}, \qquad \langle w \rangle = 1, \tag{2}$$

computed separately per population and per ASTRA iteration. The weighted 2PCF is the standard Landy–Szalay estimator with these weights on the tracer(s) and unit weights on the geometry randoms G:

$$\hat{\xi}_w(s, \mu) = \frac{D_w D_w - 2\, D_w G + GG}{GG} \tag{3}$$

for an auto-correlation, and the analogous cross form $\hat{\xi}_w = (D_w A_w - D_w G - A_w G + GG)/GG$ for the data$\times$ASTRA-random cross, where A_w denotes the weighted ASTRA randoms. Pair counts are normalised by the sums of weights, which `pycorr` handles natively. The geometry-random pair count GG is computed once and reused by every estimator. The multipoles $\xi_{w,\ell}(s)$ for $\ell = 0, 2$ follow by the usual Legendre projection.

The mean normalisation (2) has a useful consequence: the *uniform* scheme $f \equiv 1$ gives $w_i \equiv 1$ and (3) reduces to the ordinary 2PCF. We use this as a built-in unit test (the uniform weighted curve must coincide with the standard full-data ξ).

3.3 Marked correlation for signed marks

For the signed mark $m = r$ (which changes sign between voids and knots), the mean-normalised weighted estimator measures the marked-correlation *numerator* $W(s)$. The marked correlation function itself is the ratio

$$M(s) = \frac{1 + W(s)}{1 + \xi(s)}, \tag{4}$$

with ξ the unmarked (uniform) correlation; $M(s) = 1$ is the null of no mark–clustering correlation, $M > 1$ that high-r objects cluster more strongly at scale s. We report $M_0(s)$ formed per iteration from the saved W_0 (signed) and ξ_0 (uniform) arrays.

4 Weight schemes

We measure the following $f(r)$ (Table 1). All but the signed mark are non-negative, so their weighted $\xi_{w,\ell}$ are clean Landy–Szalay objects; the signed mark is reported through the marked-correlation ratio (4).

Scheme	$f(r)$	character
uniform	1	baseline; reproduces the standard 2PCF (unit test)
knot	$(1+r)/2$	overdense emphasis (continuous Q4)
void	$(1-r)/2$	underdense emphasis (continuous Q1)
rank	$\mathrm{rank}(r)/N$	distribution-robust continuous quantile
power	$[(1+r)/2]^2$	sharpened knot emphasis
exp	$\exp(2r)$	smooth, tunable, positive
signed	r	signed density weighting; marked correlation M_0

Table 1: Weight schemes. The rank is computed within each population (data, ASTRA randoms) separately, per iteration.

5 Statistics

For each scheme we measure the weighted multipoles $\ell = 0, 2$ of three statistics:

- **data auto**: data weighted by $f(r_{\mathrm{data}})$;
- **ASTRA-random auto**: ASTRA randoms weighted by $f(r_{\mathrm{A}})$;
- **data × ASTRA-random cross**: the two weighted fields of the *same* iteration, cross-correlated.

The cross quadrupole is expected to be the most informative of the three: the data carry RSD while the ASTRA randoms do not, so the cross $\ell = 2$ probes the data's velocity anisotropy modulated by the environment weighting, whereas the ASTRA-random auto $\ell = 2$ is expected to be weak (those points have no peculiar velocity).

6 Implementation

`scripts/pipeline_fullbox_weighted.py` runs the full $2\,h^{-1}\,\mathrm{Mpc\,Gpc}$ box. Per ASTRA iteration it (i) generates the ASTRA randoms, (ii) builds the joint catalog and runs `astra.classify_fast` to obtain r for every object, (iii) aligns r to the data and ASTRA-random position arrays and *stores* them (`fbw_rvalues_iter{it}.npz`, `float32`) so the weighting can be recomputed or extended without re-running the (expensive) Delaunay, and (iv) computes the weighted multipoles of the three statistics for all seven schemes. Outputs go to `data/fullbox_weighted/{cosmo}_hod{NNN}/` with prefix `fbw_`. `scripts/plot_fullbox_weighted.py` produces the figures; `queue/run_fullbox_weighted.sh` is the CPU-node wrapper. The `pycorr` weighted API and the uniform-equals-unweighted identity were validated in a smoke test before the first run.

7 First test

- Catalog: `c000/hod484` (the Fisher/emulator fiducial), full box.
- ASTRA iterations: 3; display as iteration mean ± std.
- Binning: $s \in [0, 150]\, h^{-1}\,\mathrm{Mpc}$, 15 bins; μ in 240 bins; $\ell = 0, 2$.
- Schemes: all seven of Table 1.

8 Results

(To be populated when the first run 55159247 completes.) The planned figures are

- `weighted_xi_multipoles.png`: $s^2 \xi_{w,\ell}(s)$ for the positive schemes, 3×2 (data / ASTRA-random / cross $\times\ \ell = 0, 2$);
- `marked_correlation_signed.png`: $M_0(s)$ for the signed mark, per statistic.

Key checks to record: (i) the uniform curve coincides with the standard full-data ξ; (ii) the void and knot schemes bracket the uniform curve as the underdense/overdense-emphasised correlations; (iii) the cross quadrupole carries RSD signal while the ASTRA-random auto quadrupole is weak; (iv) $M_0(s)$ departs from unity where the environment correlates with clustering.

9 Outlook

Once validated, the weighted vectors can be dropped into the existing Fisher/emulability machinery for a like-for-like comparison against the quantile legs: do the continuous weights carry the same (or more) cosmological information at a lower data-vector dimension, and are they more emulable (smoother in the parameters)? Natural extensions are tunable exponents (the `power`/`exp` knobs), cross-weightings (void-weighted × knot-weighted, the continuous analog of the void/knot cross legs that were the Fisher workhorses), and the standardised zero-mean mark (the mark variance / density-fluctuation weighting).

References

[1] R. K. Sheth, "Marked correlations," MNRAS **364** (2005) 796.

[2] M. White, "Marked correlations and optimal weights," JCAP **11** (2016) 057.

[3] E. Massara et al., "Using the marked power spectrum to detect the signature of neutrinos," PRL **126** (2021) 011301.